

Submission A solution to Problem 1

A computably AUT-countable example with no finite existential automorphism basis

Theorem 1. *There is a computably represented countable structure $\mathcal{A}$ such that every automorphism of every copy of $\mathcal{A}$ is computable from that copy, and hence $\mathcal{A}$ is computably AUT-countable on a cone, but for every finite parameter set $P \subseteq A$ there is an automorphism π of $\mathcal{A}$ whose graph is not Σ_1^{in} -definable over $P \cup \pi(P)$.*

Proof. Let $V = [\omega]^{<\omega}$, with addition $+$ given by symmetric difference; thus V is a countably infinite vector space over $\mathbb{F}_2$. Fix a computable sequence $(D_i)_{i \in \omega}$ of finite subsets of V such that every finite subset of V occurs cofinally often.

We use a one-sorted computable coding of two sorts. The domain is the computable disjoint union

$$(\{0\} \times \omega) \sqcup (\{1\} \times V),$$

identified effectively with ω . We write its two unary predicates as I and S . The language also has a binary relation $<$ and a ternary relation R . On I the relation $<$ is the usual order of ω , and $<$ is false if one of its arguments is outside I . For $i \in I$ and $x, y \in S$, put

$$R(i, x, y) \iff x + y \in D_i,$$

where we identify elements of the S -sort with vectors in V ; on all other triples R is false. This is a computable relational structure.

First compute its automorphism group. Since $(I, <)$ has order type ω , every automorphism fixes I pointwise. If $f \in \text{Aut}(\mathcal{A})$, let $c = f(0_V)$. For every $h \in V$ choose i with $D_i = \{h\}$. Then $R(i, x, x+h)$ for all $x \in V$, and preservation of R gives $f(x+h) = f(x) + h$. Taking $x = 0_V$ gives

$$f(h) = c + h \quad (h \in V).$$

Conversely, for each $c \in V$, the map τ_c which fixes I pointwise and sends $x \in S$ to $x + c$ preserves all differences $x + y$, hence preserves R . Thus

$$\text{Aut}(\mathcal{A}) = \{\tau_c : c \in V\}.$$

This already implies computable AUT-countability, with no cone oracle. Let $\mathcal{B} \cong \mathcal{A}$, and let $\sigma \in \text{Aut}(\mathcal{B})$. Transporting σ to $\mathcal{A}$, it is some translation τ_c . Choose an element i of the

I -sort of $\mathcal{B}$ corresponding to some index with $D_i = \{c\}$. A Turing program with oracle the atomic diagram of $\mathcal{B}$ and with this natural number i hardwired computes σ : it queries the sort predicate, outputs n on inputs n in the I -sort, and on inputs x in the S -sort it searches for the unique y with $R(i, x, y)$. Hence every automorphism of every copy is computable relative to the copy.

It remains to prove the promised failure of Σ_1^{in} -definability. Fix a finite parameter set $P \subseteq A$. Let

$$M = \max(P \cap I)$$

if $P \cap I \neq \emptyset$, and put $M = -1$ otherwise. Let

$$W = \text{span}\left(\bigcup_{i \leq M} D_i\right) \leq V$$

(where $W = \{0\}$ if $M = -1$). The subspace W is finite. Choose a nonzero $h \in V \setminus W$, and let $\pi = \tau_h$. Put $Q = P \cup \pi(P)$. Since the quotient V/W is infinite and $Q \cap S$ is finite, choose $x \in V$ such that the two cosets

$$x + W, \quad x + h + W$$

are disjoint from $Q \cap S$. Let $y = x + h$.

We shall show that the graph-pair (x, y) is not isolated inside $\text{graph}(\pi)$ by any existential formula over Q . This suffices, because if a countable disjunction of existential formulas defined exactly the graph, then any disjunct true of (x, y) would itself have to be contained in the graph.

Let

$$\exists \bar{z} \theta(\bar{q}, u, v, \bar{z})$$

be an existential formula with parameters $\bar{q}$ from Q , and suppose it is true at $(u, v) = (x, y)$. Choose witnesses. Passing to one true conjunction of literals in a disjunctive normal form of the quantifier free part, it is enough to preserve the truth values of the finitely many literals in that conjunction.

Call an I -sort value *low* if it is at most M , and *high* otherwise. All low markers have their associated finite sets D_i contained in W . Let F be the finite set of all S -sort elements occurring among x, y , the parameters, and the chosen witnesses, and put $C = y + W$. We shall add a vector t precisely to those members of F which lie in C . The coset C contains no parameter and does not contain x .

Choose $t \in V$ outside the following finite forbidden set. First avoid $t = 0$ and all choices causing a shifted element of $F \cap C$ to collide with an unshifted element of $F \setminus C$. Also, for every $i \leq M$, every $a \in F \cap C$, every $b \in F \setminus C$, and every $d \in D_i$, avoid $t = a + b + d$. These exclusions guarantee that all equalities and all R -literals whose first argument is a low marker are preserved: within C all differences are unchanged; outside C nothing is changed; and across the cut the original difference is not in W , while the new difference is kept out of each low D_i .

There are also finitely many high I -sort equality classes among the chosen witnesses. For each such class μ , and for each pair consisting of a positive and a negative R -literal with first argument in μ and other two arguments in the S -sort, we additionally require that the two resulting differences after the vector shift be unequal. If the two differences are shifted in the same way this inequality already follows from the original true diagram; otherwise it excludes one further value of t . Thus only finitely many more values are forbidden, and we choose t avoiding all of them.

Now perform the vector shift by this t . For each high equality class μ , let A_μ be the finite set of new differences required by the positive R -literals with first argument in μ , and let B_μ be the finite set of new differences forbidden by the negative such literals. By the choice of t , $A_\mu \cap B_\mu = \emptyset$. Replace the high I -sort witnesses by new indices, all above M and in the same equality and order pattern as before, with associated sets exactly A_μ . This is possible because each finite subset of V occurs cofinally often among the D_i 's. Sort predicates, equalities, and $<$ -literals are preserved by construction. Literals involving R with an argument in the wrong sort remain false, low-marker R -literals were handled above, and high-marker R -literals are made true or false according as their resulting difference lies in A_μ or not.

Consequently the same existential formula is true of the new pair

$$(x, y + t) = (x, x + h + t).$$

But $t \neq 0$, so $y + t \neq y = \pi(x)$. Hence this new pair is not in the graph of π . Therefore no existential formula over $Q = P \cup \pi(P)$ which is true of $(x, \pi(x))$ can be contained in the graph of π , and the graph of π is not Σ_1^{in} -definable over Q .

Since P was arbitrary, the required property holds for every finite parameter set. $\square$